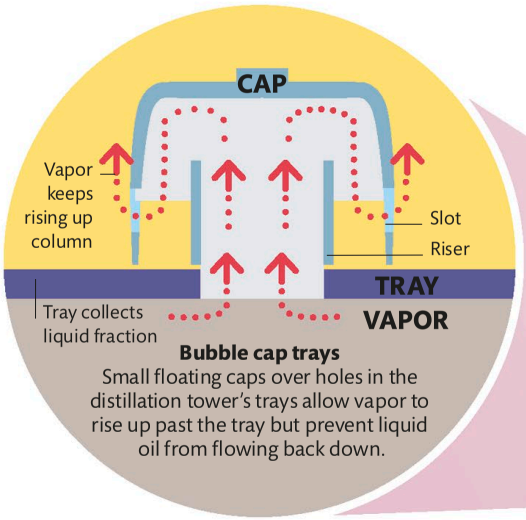
Oil refineries

Crude oil is extracted from oil deposits in Earth's crust and shipped or piped to refineries. It consists of a combination of many types of hydrocarbons. These can be separated out into various products that are used in different ways.

Fractional distillation

The different hydrocarbons in crude oil have varying boiling points. This means that they can be separated out by evaporating the oil then condensing the gases into different products at different temperatures. This takes place in a distillation tower. Substances with lower boiling points condense higher up the tower. Trays at calibrated heights collect the substances, known as fractions.



Bubble cap trays
Small floating caps over holes in the distillation tower's trays allow vapor to rise up past the tray but prevent liquid oil from flowing back down.

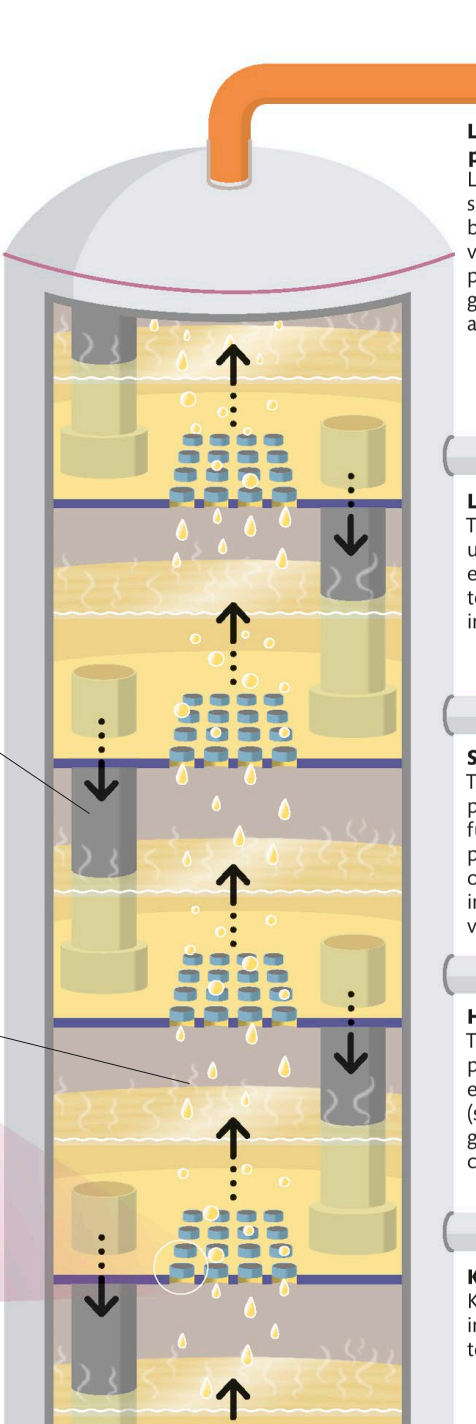
5 Tray collection
At each level in the column, as a fraction of the oil vapor cools and condenses to a liquid, it is collected on a tray and piped away for processing and storage.

Pipe called a downcomer channels liquid down from one tray to another

4 Vapor rises
Fractions of hydrocarbons with lower boiling points continue to travel up the column further than heavier fractions, passing through holes in trays as they rise.

Vapor rises, passing through holes in trays

3 Distillation
At a certain height and temperature within the tower, a fraction condenses into liquid, separating out from the rest of the oil vapor, which rises up the column.



Liquid petroleum gas
Lighter hydrocarbons, such as propane and butane, remain as vapor. These are processed into bottled gases used in heating and cooking.

Light naphtha
This fraction is often used to produce ethylene, which is used to make many plastics, including polyethylene.

Straight-run gasoline
This is gasoline produced without further chemical processing. Almost half of crude oil is refined into gasoline used as vehicle fuel.

Heavy naphtha
This fraction is often processed further, for example, by cracking (see below), to produce gasoline and other crude-oil products.

Kerosene
Kerosene is used as a fuel in heaters or refined further to produce potent jet fuel.